

Low-Light Image Enhancement and Downstream Object Detection

A Comparative Evaluation of Zero-DCE, Zero-DCE++, Classical and Hybrid Enhancement with YOLOv8n on the ExDark Dataset

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Abstract

Project 18 asks whether low-light image enhancement improves downstream recognition rather than only perceptual quality. We evaluate a two-stage enhancement–detection pipeline on a processed ExDark dataset containing 7,345 images and 12 object classes. The comparison uses a raw baseline and five enhancement cases: CLAHE, CLAHE with bilateral filtering, Zero-DCE trained from scratch, pretrained Zero-DCE++, and Zero-DCE++ with bilateral filtering. All trained YOLOv8n detectors use a 40-epoch budget and the same data split and hyperparameters. The scratch Zero-DCE main result is a direct cascade that keeps the raw-domain detector fixed; it obtains 22.91% mAP@0.5 and 9.70% mAP@0.5:0.95. The separately consolidated raw baseline obtains 61.46% and 28.62%, while Zero-DCE++–bilateral reaches 62.44% and 29.21%. Because the scratch row uses a fixed detector whereas the other enhanced rows use matched fine-tuning, its degradation measures domain transfer and is not a pure enhancer-only comparison. The evidence supports a method-dependent conclusion: visual enhancement alone does not guarantee better recognition, and controlled comparison requires identical detector settings and traceable artifacts.

Keywords: low-light image enhancement; object detection; Zero-DCE; Zero-DCE++; CLAHE; bilateral filtering; YOLOv8n; ExDark; domain shift.

1 Introduction and Research Question

1.1 Background and Motivation

Object detection in low-light scenes supports applications such as nighttime surveillance and intelligent transportation. Insufficient exposure, weak local contrast, sensor noise, color distortion, and non-uniform illumination can obscure object boundaries and texture. ExDark was created to study these conditions and contains real low-light scenes from 12 object classes [1].

Enhancement is a plausible preprocessing step, but a clearer image is not necessarily a better detector input. Classical methods redistribute local intensity, whereas Zero-Reference Deep Curve Estimation (Zero-DCE) learns spatially varying transformations without paired bright reference images [2]. Both approaches can also change color, noise, and texture statistics, causing a mismatch with a detector trained on raw images. Project 18 must therefore evaluate enhancement quality and downstream recognition as a coupled system.

1.2 Study Design

The main comparison uses a **raw dark baseline** and five enhancement cases while preserving image geometry and labels: **CLAHE**, **CLAHE + bilateral filtering**, **Zero-DCE trained from scratch**, **pretrained Zero-DCE++**, and **Zero-DCE++ + bilateral filtering**. The scratch model is one of the five enhancement cases, but it is not interchangeable with the pretrained Zero-DCE++ model.

Each variant is passed to YOLOv8n. A fixed-detector cascade tests direct transfer from the raw domain, whereas matched-domain fine-tuning trains YOLOv8n on the corresponding enhanced images. Reporting these protocols separately prevents a change in preprocessing from being confused with a change in detector training.

1.3 Research Question

The central research question is:

Across raw, classical, learned, and hybrid low-light image variants, which enhancement strategies improve downstream YOLOv8n detection on ExDark, and does an improvement in visual quality consistently correspond to an improvement in recognition performance?

The raw baseline and five enhancement cases use the same ExDark split and evaluation settings. Whenever YOLOv8n is trained, its initialization and hyperparameters are held fixed. The scratch case instead reuses the raw-domain detector without additional fine-tuning, so detector protocol is reported for every row. Comparing the two Zero-DCE variants evaluates complete learned-enhancement pipelines rather than a pure architectural ablation because their architectures and checkpoint origins differ. Comparing standalone Zero-DCE++ with its bilateral variant isolates the additional denoising stage within the pretrained learned branch.

1.4 Contribution and Scope

The study provides a common evaluation structure for classical, learned, and hybrid preprocessing followed by YOLOv8n. It separates measured observations from possible explanations and limits its conclusions to the evaluated ExDark splits, checkpoints, and training configurations.

2 Related Work

2.1 Low-Light Image Enhancement

Classical enhancement methods manipulate image statistics without learning a mapping from data. Contrast-limited adaptive histogram equalization (CLAHE) improves local contrast while clipping histogram amplification [4]. In this project, CLAHE is applied only to the lightness channel in CIE LAB space so that the chromatic channels remain unchanged. Bilateral filtering combines spatial and photometric proximity to smooth noise while preserving strong edges [5]. The combination provides an interpretable non-learning baseline, but it may amplify noise or alter local texture in ways that affect a detector.

Deep enhancement methods learn a nonlinear transformation. Zero-DCE formulates enhancement as image-specific estimation of pixel-wise light-enhancement curves and uses non-reference losses

instead of paired target images [2]. Its extended version introduces Zero-DCE++, a smaller model that uses depthwise-separable convolution and reuses a compact curve map across iterations [3]. These properties make the family appropriate for ExDark, for which corresponding well-exposed reference images are unavailable.

2.2 Low-Light Enhancement for Recognition

ExDark was introduced to characterize how real low illumination affects object features and detection [1]. Its motivation is directly aligned with this project: enhancement should be assessed not only by its appearance but also by its effect on a high-level task. The present work uses YOLOv8n, the smallest YOLOv8 detection variant, initialized from the official pretrained weights and fine-tuned for the 12 ExDark classes [6]. The detector is therefore *not* trained from random initialization; only the project-specific fine-tuning starts from the released checkpoint.

2.3 Position of This Study

The report does not propose a new enhancement architecture. Instead, it performs a controlled comparison of raw, classical, learned, and hybrid inputs. Its main methodological distinction is between (i) a fixed detector evaluated after an enhancer and (ii) a detector fine-tuned on the enhanced domain. This separation tests whether enhancement transfers directly and whether detector adaptation changes the conclusion.

3 Dataset and Data Preparation

3.1 Dataset Source and Local Version

The official Exclusively Dark (ExDark) dataset contains 7,363 visible-light low-light images, ten illumination conditions, and 12 object categories [1]. The official dataset is available at <https://github.com/cs-chan/Exclusively-Dark-Image-Dataset>. The project repository identifies its intermediate YOLO export as Roboflow dataset version 12: <https://universe.roboflow.com/project-h68de/exdark-kd37x/dataset/12>. The local processed version contains 7,345 usable image-label pairs and the classes Bicycle, Boat, Bottle, Bus, Cat, Cup, Motorbike, People, Table, car, chair, and dog. The report must retain the distinction between the official size and the processed size; the 18-image difference is a data-preparation issue rather than a change to the published dataset.

3.2 Split and Label Conversion

The project uses a deterministic split with seed 42: 5,142 training images, 1,469 validation images, and 734 test images. The source export stored some annotations in a nine-field oriented-box representation. The preparation step converted 23,146 annotations to the five-field YOLO detection format (c, x_c, y_c, w, h) and discarded three zero-area boxes. Image identifiers and encoded-image hashes were used to check for duplication across the selected splits. The resulting geometry is shared by every photometric variant so that the same detection labels can be copied without coordinate changes.

Table 1 summarizes the image and label-file counts verified in the local processed dataset. Each image has one corresponding YOLO-format label file, and the three subsets preserve the intended 70%/20%/10% split.

Table 1: Split of the 7,345 images in the processed ExDark benchmark.

Split	Images	Label files	Share (%)	Folder in the project
Train	5,142	5,142	70.0	Dataset/exdark_yolo_dark/train/
Validation	1,469	1,469	20.0	Dataset/exdark_yolo_dark/valid/
Test	734	734	10.0	Dataset/exdark_yolo_dark/test/
Total benchmark	7,345	7,345	100.0	Dataset/exdark_yolo_dark/

Table 2 reports the exact class-ID mapping stored in the processed dataset configuration. The original capitalization is retained because these strings are the display names associated with the numerical class indices; the detector itself is trained and evaluated using IDs 0–11. The final column describes typical visual characteristics that may affect recognition in nighttime scenes; it is provided as qualitative context rather than as an additional dataset annotation.

Table 2: The 12 object classes in the processed ExDark dataset and their typical appearance in nighttime scenes.

ID	Class	Typical appearance in ExDark nighttime images
0	Bicycle	Thin frames and spokes can merge with dark backgrounds.
1	Boat	Hull boundaries may be weakened by darkness and reflections on water.
2	Bottle	Small reflective highlights may be more visible than the full object boundary.
3	Bus	A large silhouette is often accompanied by bright windows or vehicle lights.
4	Cat	A small body may be difficult to resolve, while the eyes can remain conspicuous.
5	Cup	Small cups or glasses may have weak contrast on tables in nighttime scenes.
6	Motorbike	Thin structural parts may be obscured, while headlights can produce strong local contrast.
7	People	Body boundaries may blur because of low contrast, occlusion, or motion.
8	Table	Tabletops and legs may blend into surrounding outdoor dining scenes.
9	Car	Vehicle contours may be obscured by darkness, glare, or bright headlights.
10	Chair	Thin legs and backrests can be difficult to distinguish in dark street scenes.
11	Dog	Dark coats and small body regions may blend into the nighttime background.

3.3 Photometric Dataset Variants

The main comparison considers a raw baseline and five enhanced inputs: CLAHE; CLAHE followed by bilateral filtering; Zero-DCE trained from scratch; pretrained Zero-DCE++; and pretrained Zero-DCE++ followed by bilateral filtering. CLAHE uses a clip limit of 2.0 and an 8×8 tile grid. The classical CLAHE–bilateral implementation uses a neighborhood diameter of 7 with $\sigma_{\text{color}} = \sigma_{\text{space}} = 50$. The currently recorded Zero-DCE++–bilateral dataset instead uses diameter 5 and both sigma values equal to 25. These two filter configurations must not be described as the same experimental case.

All transformations modify pixel values only. The main test matrix uses the same train, validation, and test membership, the same bounding-box annotations, and the same detector settings across variants. The exact manifests, enhancement checkpoints, and evaluation commands should still be archived with the submitted results to make this controlled protocol independently auditable.

4 Methods

4.1 Raw Baseline and Five Enhancement Scenarios

Let I denote a raw low-light image, E_k an image transformation, and D_k a YOLOv8n detector. The raw baseline and five enhancement transformations are

$$\begin{aligned} E_0(I) &= I, & E_1(I) &= E_{\text{CLAHE}}(I), \\ E_2(I) &= B_C(E_{\text{CLAHE}}(I)), & E_3(I) &= E_{\text{scratch}}(I), \\ E_4(I) &= E_{\text{ZDCE}++}(I), & E_5(I) &= B_Z(E_{\text{ZDCE}++}(I)), \end{aligned} \quad (1)$$

where B_C and B_Z are bilateral filters with the branch-specific parameters defined below. For the raw baseline and matched-domain cases, the prediction is

$$\hat{Y}_k = D_k(E_k(I)), \quad k \in \{0, 1, 2, 4, 5\}. \quad (2)$$

For the scratch Zero-DCE direct cascade, the raw-domain detector is held fixed:

$$\hat{Y}_3 = D_0(E_3(I)). \quad (3)$$

Thus, the main result table contains six rows: the raw baseline and five enhancement cases. The detector protocol is stated explicitly because the scratch row is a fixed-detector transfer test.

4.2 Classical Baseline: CLAHE and Bilateral Filtering

Each BGR input is converted to CIE LAB. CLAHE is applied only to the L channel with a clip limit of 2.0 and an 8×8 tile grid [4]. The original a and b channels are then recombined and the image is converted back to BGR. For E_2 , the CLAHE output is processed by B_C , an OpenCV bilateral filter with diameter 7 and $\sigma_{\text{color}} = \sigma_{\text{space}} = 50$. Comparing E_1 with E_2 isolates the effect of the bilateral denoising stage within the classical branch. For an input image X , both bilateral branches implement the edge-preserving weighted average

$$\begin{aligned} B(X)(p) &= \frac{1}{W_p} \sum_{q \in \Omega_p} \exp\left(-\frac{\|p - q\|_2^2}{2\sigma_{\text{space}}^2}\right) \\ &\quad \times \exp\left(-\frac{\|X(p) - X(q)\|_2^2}{2\sigma_{\text{color}}^2}\right) X(q). \end{aligned} \quad (4)$$

where W_p is the sum of the two Gaussian weights over the neighborhood Ω_p [5]. The equation is common to B_C and B_Z ; only their recorded diameter and standard-deviation parameters differ.

4.3 Pretrained Zero-DCE++ and Hybrid Denoising

Zero-DCE++ replaces standard convolutions with depthwise-separable operations and predicts a compact curve map that is reused across eight enhancement iterations [3]. In the checkpoint-compatible implementation, the predicted RGB map $r \in [-1, 1]^{3 \times H \times W}$ is applied as

$$\begin{aligned} z_0 &= I, \\ z_n &= z_{n-1} + r \odot (z_{n-1}^2 - z_{n-1}) \\ &= z_{n-1} + A \odot z_{n-1} \odot (1 - z_{n-1}), \quad A = -r, \quad n = 1, \dots, 8. \end{aligned} \quad (5)$$

where $\odot$ denotes element-wise multiplication. Writing $A = -r$ makes the repository’s checkpoint sign convention explicit and recovers the LE-curve form used in the paper. Unlike the scratch

model below, the same three-channel map is reused at every iteration. Both E_4 and E_5 use the same official pretrained Zero-DCE++ checkpoint. The standalone case E_4 sends its output directly to the matched-domain detector. The hybrid case E_5 additionally applies B_Z , an OpenCV bilateral filter with diameter 5 and $\sigma_{\text{color}} = \sigma_{\text{space}} = 25$. Because B_C and B_Z use different parameters, their effects are interpreted only within their respective branches.

4.4 Zero-DCE Trained from Scratch

The local DCE-Net contains seven 3×3 convolutional layers, 32 feature channels in the first six layers, symmetric skip concatenations, and 24 output maps, for a total of 79,416 trainable parameters. The final tanh activation restricts these maps to $[-1, 1]$. They are divided into eight RGB triplets $A_n \in [-1, 1]^{3 \times H \times W}$, one for each enhancement iteration. Given a normalized input $x_0 = I \in [0, 1]^{3 \times H \times W}$, the local implementation computes

$$x_n = x_{n-1} + A_n \odot x_{n-1} \odot (1 - x_{n-1}), \quad n = 1, \dots, 8, \quad (6)$$

and clamps x_8 to $[0, 1]$ as a numerical safeguard. This is the higher-order LE-curve construction introduced by Zero-DCE [2], but here all parameters and the checkpoint are learned from the project data rather than loaded from a pretrained model.

The network is trained without paired reference images. To state the executed objective precisely, let $Y = x_8$, let $P_i(X)$ be the RGB-mean intensity averaged over the i -th non-overlapping 4×4 patch, and let Δ_d denote a directional difference for $d \in \mathcal{D} = \{\text{left, right, up, down}\}$. Let N_p be the number of pooled spatial positions across the minibatch. The spatial-consistency term is

$$\mathcal{L}_{\text{spa}} = \frac{1}{N_p} \sum_{i=1}^{N_p} \sum_{d \in \mathcal{D}} [\Delta_d P_i(Y) - \Delta_d P_i(I)]^2. \quad (7)$$

If $\bar{Y}_k$ is the mean of all RGB values in the k -th non-overlapping 16×16 patch, the exposure-control term is

$$\mathcal{L}_{\text{exp}} = \frac{1}{M} \sum_{k=1}^M |\bar{Y}_k - 0.6|. \quad (8)$$

For sample b , let μ_b^c be the spatial mean of channel $c \in \{R, G, B\}$. The implemented color-constancy term is

$$\begin{aligned} \mathcal{L}_{\text{col}} = \frac{1}{B} \sum_{b=1}^B [& (\mu_b^R - \mu_b^G)^2 + (\mu_b^R - \mu_b^B)^2 \\ & + (\mu_b^G - \mu_b^B)^2 + \varepsilon]^{1/2}, \quad \varepsilon = 10^{-8}. \end{aligned} \quad (9)$$

Finally, for the complete parameter tensor $A \in \mathbb{R}^{B \times C \times H \times W}$, where $C = 24$, the illumination-smoothness term is the normalized squared total variation

$$\mathcal{L}_{\text{tv}} = \frac{1}{BC} \left(\frac{\|\nabla_h A\|_F^2}{(H-1)W} + \frac{\|\nabla_w A\|_F^2}{H(W-1)} \right). \quad (10)$$

The resulting local objective is

$$\mathcal{L} = \mathcal{L}_{\text{spa}} + 10\mathcal{L}_{\text{exp}} + 5\mathcal{L}_{\text{col}} + 200\mathcal{L}_{\text{tv}}, \quad (11)$$

so the reported equations capture the repository implementation rather than silently replacing it with the paper's more general notation. The four terms respectively preserve local spatial differences, control exposure, penalize global color imbalance, and encourage smooth curve-parameter maps. In the local implementation, RGB inputs are resized to 256×256 , and each epoch processes at most 200 randomly selected training images. The documented command runs 10 epochs with batch size 8. Optimization uses Adam with learning rate 10^{-4} and weight decay 10^{-4} . No random seed or validation loop is recorded for this enhancer run; the saved checkpoint is selected by the lowest average training loss.

4.5 Generation of Enhanced Dataset Variants

The learned enhancers are fully convolutional, so dataset generation processes each image at its original spatial resolution. Pixel values are normalized to $[0, 1]$, enhanced under inference mode, clamped to the valid range, converted back to BGR, and quantized to 8-bit values. The pretrained-data preparation pipeline stores enhanced outputs as lossless PNG and verifies their shapes. The scratch Zero-DCE pipeline instead preserves each source filename and extension when calling OpenCV’s image writer; consequently, a JPEG source can undergo JPEG re-encoding. This encoding difference is retained here because it is part of the executed pipeline and should not be described as lossless. Both pipelines copy the normalized YOLO label files without coordinate changes. Therefore, split membership, geometry, class indices, and bounding boxes remain fixed, although image appearance and, for the scratch branch, file encoding may change.

4.6 Downstream Detector

YOLOv8n is initialized from `yolov8n.pt` and fine-tuned for 12 classes at an input size of 640×640 . YOLOv8 uses a multi-scale backbone and neck with an anchor-free detection head [6]. The recorded raw and scratch-domain detector artifacts both use 40 epochs, batch size 16, image size 640, seed 0, deterministic execution, automatic optimizer selection, and mosaic disabled for the final ten epochs. Their augmentation is not identical: the raw run records `hsv_v` = 0.4, whereas the scratch-domain adapted run records `hsv_v` = 0.1. The scratch Zero-DCE main case does not introduce another YOLO training run; its enhanced test images are evaluated with the fixed raw-domain detector. The scratch-domain detector is treated only as a domain-adaptation ablation.

The phrase *trained from scratch* applies only to the Zero-DCE enhancement network described in Section 4.4. YOLOv8n is never trained from random initialization. The enhancer’s optimization schedule must therefore not be reported as the detector retraining schedule.

4.7 Comparison Strategy

The raw case E_0 establishes the downstream baseline. Comparing E_1 with E_2 isolates bilateral filtering in the classical branch, while comparing E_4 with E_5 isolates it in the pretrained learned branch. The scratch case E_3 measures direct transfer through a fixed detector. Its score is therefore not interpreted as a controlled enhancer-only comparison against matched-domain rows. A comparison is treated as controlled only when image identifiers, labels, split, detector protocol, epoch budget, augmentation, and evaluation options are verified to match.

5 Experimental Setup

5.1 Setup 1: Raw Baseline and Five Enhancement Cases

The main matrix contains raw, CLAHE, CLAHE–bilateral, scratch Zero-DCE, pretrained Zero-DCE++, and Zero-DCE++–bilateral inputs. The detector runs use a 40-epoch budget and 640×640 inputs, but their exported arguments must be reported per run rather than assumed identical. The scratch case is evaluated as a direct cascade with the recorded raw-domain detector; the other enhanced rows use their recorded matched-domain detectors. Precision, Recall, mAP@0.5, and mAP@0.5:0.95 are reported on the held-out test set, with the detector protocol stated explicitly.

5.2 Setup 2: Detector-Domain Alignment Ablation

The scratch Zero-DCE main row already uses the fixed raw-domain detector. A separately fine-tuned scratch-domain detector may be reported as a domain-adaptation ablation, but it does not replace the direct-cascade result or create another enhancement case.

5.3 Setup 3: Analysis and Ablations

Three analyses are required. First, CLAHE is compared with CLAHE–bilateral. Second, standalone Zero-DCE++ is compared with Zero-DCE++–bilateral to isolate explicit denoising. Third, per-class and qualitative analyses determine whether changes are concentrated in small, low-contrast, or partially occluded objects. The scratch direct cascade and its optional matched-domain ablation must be interpreted separately because they use different detector protocols.

5.4 Metrics and Reporting Rules

Downstream performance is reported as class-averaged Precision, Recall, mAP@0.5, and mAP@0.5:0.95. The final metric averages AP across IoU thresholds from 0.5 to 0.95 and is therefore stricter than mAP@0.5 [7]. All values in this report are written as percentages for readability. Differences between two percentage-valued metrics are reported in percentage points, not as relative percentages.

Because ExDark has no paired well-exposed targets, PSNR and SSIM against a ground-truth bright image are not applicable. Any no-reference image-quality metric must use its standard published implementation and version. The heuristic functions currently named NIQE and BRISQUE in the repository are proxies and are not reported as standard NIQE or BRISQUE scores.

5.5 Recorded Configuration Summary

Table 3 summarizes the two trainable stages. The main comparison fixes the detector-training hyperparameters across all trained runs; the scratch main row reuses the raw detector. The exact exported argument file should be preserved with the final result package. The Zero-DCE++ enhancer uses an official pretrained checkpoint and is not retrained for this study.

Table 3: Training settings for the principal experiment blocks.

Run block	Epochs	Batch	Image size	Seed	hsv_v	Init.
Zero-DCE scratch enhancer	40	8	256	42	–	random
YOLOv8n training runs	40	16	640	42	identical	yolov8n.pt

6 Results and Discussion

6.1 Main Held-Out Test Comparison

Table 4 reports the raw baseline and five enhancement cases. The scratch Zero-DCE row is the direct-cascade result from the recorded scratch experiment: the 40-epoch raw-domain YOLOv8n detector is kept fixed and evaluated on scratch-enhanced images. It is therefore labelled separately from the other enhanced rows, which use their recorded matched-domain detectors.

Table 4: Held-out results for the raw baseline and five enhancement cases.

Input domain	Precision (%)	Recall (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)
Raw dark	66.71	58.83	61.46	28.62
CLAHE	65.43	59.35	61.88	28.72
CLAHE + bilateral	65.72	58.00	61.61	28.56
Zero-DCE scratch, fixed raw YOLO [†]	41.31	24.50	22.91	9.70
Pretrained Zero-DCE++	67.71	52.81	58.81	27.05
Zero-DCE++ + bilateral	70.40	56.77	62.44	29.21

[†]This row is a fixed-detector direct cascade. Its experiment-level summary reports evaluation on the test split, but a separate `measured_metrics.json` has not yet been archived. It must not be interpreted as a matched-domain fine-tuning result.

Relative to raw input, CLAHE changes mAP@0.5 and mAP@0.5:0.95 by +0.42 and +0.10 percentage points. CLAHE-bilateral changes them by +0.15 and -0.06 points. Standalone Zero-DCE++ changes the two AP measures by -2.65 and -1.57 points. Zero-DCE++-bilateral changes precision, recall, mAP@0.5, and mAP@0.5:0.95 by +3.69, -2.06, +0.98, and +0.59 points, respectively.

The scratch direct cascade is interpreted against its paired raw-domain result from the same scratch experiment, not against the separately consolidated raw row in Table 4. In that paired block, mAP@0.5 decreases from 62.35% to 22.91%, a drop of 39.44 percentage points. Because the detector is held fixed, this result measures a severe input-domain transfer penalty. It does not measure the performance attainable after adapting YOLOv8n to scratch-enhanced training images.

6.2 Forty-Epoch Validation Logs

Table 5 records the final validation row from each available 40-epoch log. These values provide secondary evidence about training behavior and include auxiliary runs that are not members of the confirmed main matrix. Final method comparisons must be based on a complete set of independent held-out test results in Table 4.

Table 5: Epoch-40 validation metrics from available training logs.

Matched training/evaluation domain	Precision (%)	Recall (%)	mAP@0.5 (%)	mAP@0.5:0.95 (%)
Raw	67.75	58.47	62.29	29.13
CLAHE only	67.28	58.98	63.38	29.76
CLAHE + bilateral	69.44	58.14	62.92	29.67
Zero-DCE++ + bilateral	69.41	58.43	63.11	29.50

Within these development validation rows, the three enhanced variants have slightly higher mAP values than the raw run, but the ranking depends on the metric. CLAHE-only has the highest mAP@0.5 and mAP@0.5:0.95, whereas CLAHE-bilateral has the highest precision and lower recall than CLAHE-only. These validation values do not replace the held-out results or override the detector-protocol distinction in Table 4.

6.3 Answers to the Research Questions

The results answer the research question in a method- and protocol-dependent manner. CLAHE and CLAHE-bilateral change aggregate AP only slightly relative to raw input. Standalone Zero-

DCE++ reduces both AP measures, whereas Zero-DCE++-bilateral produces small AP gains and higher precision at the cost of lower recall relative to the raw baseline. Scratch Zero-DCE causes a large loss when inserted before the fixed raw-domain detector. This last result demonstrates a transfer mismatch, not the limit of a detector adapted to scratch-enhanced images. Visual enhancement therefore cannot be assumed to improve every recognition metric; the enhancement method, detector protocol, and selected metric must be reported together.

7 Error and Qualitative Analysis

7.1 Required Sampling Protocol

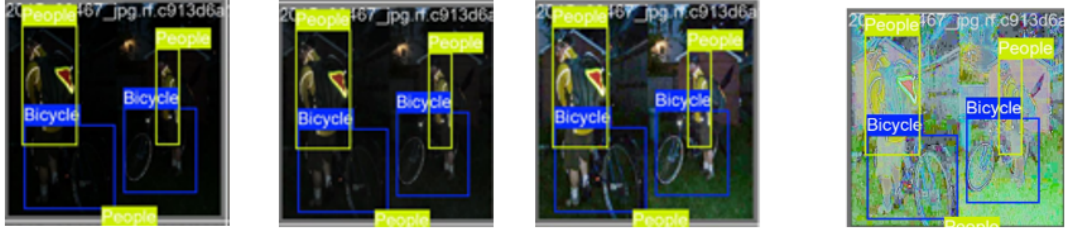
Error analysis should use the same named test images across raw, enhanced, and hybrid variants. Cases should be selected by an explicit rule—for example, the largest improvement and largest degradation in per-image detection score, plus representative low-, medium-, and high-illumination scenes—rather than only by visual attractiveness. Each figure should show the input image, the enhanced image, ground-truth boxes, predictions, class labels, and confidence scores at a shared threshold.

7.2 Error Taxonomy

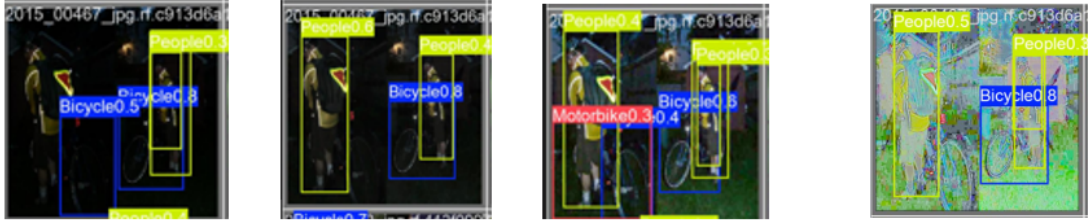
The final analysis should classify at least the following failures:

- **Missed detections:** objects remain too dark, small, occluded, or blurred for the detector to localize.
- **False positives:** amplified noise, halos, reflections, or background texture are assigned an object class.
- **Localization errors:** the class is correct but the box has insufficient overlap with the annotation.
- **Class confusion:** visually similar or poorly illuminated categories are interchanged.
- **Enhancement artifacts:** color casts, clipped highlights, over-enhancement, and over-smoothing remove useful features.

Figure 1 distinguishes two outputs from the same validation batch. Panel (a) visualizes the dataset annotations: each box and class name is a ground-truth label supplied to the evaluator, so no confidence score is attached. Panel (b) visualizes the YOLOv8n detections: each box is generated by the detector, and the number following the class name is its confidence score. A disagreement between the panels is therefore a detection error rather than a change to the dataset labels.



(a) Ground-truth class labels and bounding boxes for Raw, CLAHE, Zero-DCE++, and pretrained Zero-DCE (left to right).



(b) YOLOv8n predictions with class confidence scores for the same left-to-right method order.

Figure 1: Ground-truth annotations and detector predictions for four corresponding validation images. The upper row contains dataset labels, whereas the lower row contains model-generated boxes and confidence scores.

Panel-wise error interpretation. Reading the four prediction images from left to right gives the following qualitative observations:

- Panel 1:** The first image remains strongly underexposed. The visible detector confidences are only about 0.3–0.5, and the People instance annotated on the left is not recovered by a corresponding visible People prediction. This is a missed detection associated with weak foreground contrast.
- Panel 2:** In the second image, the left People and a Bicycle receive stronger visible scores of approximately 0.6 and 0.8. Nevertheless, the prediction set does not visibly reproduce every ground-truth box. The result therefore represents partial recovery rather than a completely correct detection.
- Panel 3:** The third image produces an additional Motorbike prediction with a confidence of approximately 0.3, although no Motorbike box exists in the corresponding ground truth. This is a false positive and likely class confusion in the overlapping person–bicycle region. The overlapping boxes also indicate localization ambiguity around adjacent objects.
- Panel 4:** The fourth image is substantially over-enhanced, with amplified contours, color shifts, and background texture. The detector still assigns visible People and Bicycle predictions, including a Bicycle score near 0.8, but it does not visibly recover all annotated instances. This illustrates that greater apparent brightness or edge strength does not guarantee higher recall and may introduce a domain shift for the detector.

Overall, enhancement appears to increase confidence for some visible objects, but it can simultaneously leave missed detections, alter box localization, or create a false class response. Because the screenshots crop part of the lower boundary and do not record the confidence threshold or verified left-to-right method names, they support error interpretation only. Exact false-positive and

false-negative counts, and any ranking of enhancement methods, must be based on the archived predictions and aggregate test metrics rather than this figure alone.

8 Conclusion and Limitations

8.1 Quantitative Findings

This project evaluates low-light enhancement as part of an enhancement–detection system rather than as an isolated visual-processing task. On the held-out ExDark test split, the effect of enhancement is method-dependent. Relative to the raw baseline, CLAHE changes $\text{mAP}@0.5$ and $\text{mAP}@0.5:0.95$ by only +0.42 and +0.10 percentage points. Adding bilateral filtering after CLAHE reduces these gains to +0.15 and -0.06 points. Standalone pretrained Zero-DCE++ decreases the two AP measures by 2.65 and 1.57 points, whereas Zero-DCE++ followed by bilateral filtering records the strongest matched-domain result: 62.44% $\text{mAP}@0.5$ and 29.21% $\text{mAP}@0.5:0.95$, improvements of 0.98 and 0.59 points over raw input. This improvement is accompanied by a 3.69-point precision gain and a 2.06-point recall decrease, so it represents a precision–recall trade-off rather than a uniform improvement in detection.

The scratch Zero-DCE direct cascade requires a separate interpretation. Within its paired scratch experiment, applying the enhancer before the fixed raw-domain detector reduces $\text{mAP}@0.5$ from 62.35% to 22.91%, a decline of 39.44 percentage points. Because YOLOv8n is not adapted to the scratch-enhanced image domain in this case, the result demonstrates severe input-domain mismatch. It does not establish that the scratch enhancer is intrinsically inferior to every matched-domain enhancement pipeline.

8.2 Qualitative Findings

The qualitative label–prediction comparison is consistent with this quantitative conclusion but does not independently rank the methods. Enhancement can increase confidence for some People and Bicycle detections, yet missed objects, imperfect localization, and an additional Motorbike false positive remain visible. The strongly enhanced panel also shows that amplified brightness, edges, and color variation do not guarantee recovery of every annotated object.

8.3 Answer to the Research Question

The project therefore answers its research question conditionally: low-light enhancement can improve downstream recognition, but the gain is small for the best evaluated matched-domain case and depends on the enhancement method, detector-domain alignment, and evaluation metric. Perceptual improvement alone is not reliable evidence of improved object detection.

8.4 Limitations

These conclusions are subject to five limitations:

- L1: Statistical uncertainty.** The detector results do not include repeated-seed experiments or confidence intervals, so run-to-run uncertainty is unknown.
- L2: Incomplete experiment provenance.** The consolidated results do not yet archive all checkpoint hashes, dataset-manifest hashes, software versions, and evaluation arguments required for complete reproducibility.

- L3: Missing standard image-quality measurement.** The study lacks a validated standard no-reference image-quality metric that could quantify the relationship between perceptual quality and detection performance.
- L4: Scratch-case evidence and protocol.** The scratch direct-cascade result is supported by the experiment-level test summary but lacks a separately archived measured-metrics file. It also uses a fixed raw detector and therefore cannot be interpreted as a matched-domain comparison.
- L5: Limited qualitative evidence.** The qualitative analysis uses only one cropped validation batch without a recorded confidence threshold or verified left-to-right method order. It cannot estimate per-class error rates or explain the aggregate test differences causally.

Repeated or bootstrapped test evaluation, complete run archives, standard image-quality measurement, and named test examples generated at a common confidence threshold would strengthen the final claims.

References

- [1] Y. P. Loh and C. S. Chan, “Getting to Know Low-Light Images with the Exclusively Dark Dataset,” *Computer Vision and Image Understanding*, vol. 178, pp. 30–42, 2019, doi: <https://doi.org/10.1016/j.cviu.2018.10.010>.
- [2] C. Guo, C. Li, J. Guo, C. C. Loy, J. Hou, S. Kwong, and R. Cong, “Zero-Reference Deep Curve Estimation for Low-Light Image Enhancement,” in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition*, pp. 1777–1786, 2020, doi: <https://doi.org/10.1109/CVPR42600.2020.00185>.
- [3] C. Li, C. Guo, and C. C. Loy, “Learning to Enhance Low-Light Image via Zero-Reference Deep Curve Estimation,” *IEEE Transactions on Pattern Analysis and Machine Intelligence*, vol. 44, no. 8, pp. 4225–4238, 2022, doi: <https://doi.org/10.1109/TPAMI.2021.3063604>.
- [4] K. J. Zuiderveld, “Contrast Limited Adaptive Histogram Equalization,” in *Graphics Gems IV*, P. S. Heckbert, Ed. San Diego, CA, USA: Academic Press, pp. 474–485, 1994, doi: <https://doi.org/10.1016/B978-0-12-336156-1.50061-6>.
- [5] C. Tomasi and R. Manduchi, “Bilateral Filtering for Gray and Color Images,” in *Proceedings of the Sixth International Conference on Computer Vision*, pp. 839–846, 1998, doi: <https://doi.org/10.1109/ICCV.1998.710815>.
- [6] G. Jocher, J. Qiu, and A. Chaurasia, *Ultralytics YOLO*, version 8.0.0, computer software, Jan. 10, 2023. Available: <https://github.com/ultralytics/ultralytics>.
- [7] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick, “Microsoft COCO: Common Objects in Context,” in *Computer Vision—ECCV 2014*, Lecture Notes in Computer Science, vol. 8693, pp. 740–755, Springer, 2014, doi: https://doi.org/10.1007/978-3-319-10602-1_48.

A Member Contributions

Table 6 records the verified role and contribution of each group member.

Table 6: Member contribution table. Replace every placeholder with verified work completed by the named member.

Member	Student ID	Role	Task
Nguyen Minh Hoang	2410315	Team Lead	Overseeing pipeline integration and overall system evaluation
Nguyen Minh Luong	23BA14184	Member	Training Zero-DCE from scratch
Le Hien Anh	2410042	Member	Training Zero-DCE++ with YOLO
Luong Quynh Anh	23BA14008	Member	Training raw model baseline and analyzing errors
Le Ba Tiep	23BA14282	Member	Training Zero-DCE++ with image noise reduction
Le Ba Ninh	23BA14224	Member	Visualize experimental data and building demo interface
Nguyen Duy Nam	23BA14205	Member	Training CLAHE with YOLO

B Reproducibility and Evidence Map

The final submission should map every table row to a data manifest, enhancement configuration, detector checkpoint, training configuration, and measured metric file. The accompanying file `REPORT_RESULTS_CHECKLIST.md` lists the remaining measurements and verification steps. The required repository-level `DATA.md` should additionally record the official dataset URL, source version, split, preprocessing procedure, and commands used to reproduce every processed dataset.